EndoMAE: A Foundation Model for Endoscopic Image Analysis via Self-Supervised Learning

Dipika Boro, Qilei Chen, Xiaolong Liang, Yu Cao, Benyuan Liu
Miner School of Computer & Information Sciences, University of Massachusetts Lowell, MA, USA

INTRODUCTION

- Early detection through endoscopic screening significantly improves patient outcomes in Colorectal cancer (CRC).
- Endoscopy produces large volumes of video data, but costly expert annotation and privacy concerns lead to scarcity of labeled datasets.
- Supervised learning approaches rely on labeled data, constraining the development of robust medical imaging models.
- Foundation models pretrained on natural images fail to capture endoscopic domain characteristics.

EndoMAE: A Self-Supervised Learning (SSL) based foundation model for endoscopic image analysis, that serves as a general-purpose backbone for downstream analysis tasks.

DATASETS

Table 1: Datasets for pretraining & downstream finetuning phases.

Phase	Dataset	Frames
Pretraining	CholecT45	90,444
	LDPolyp	40,186
	Colonoscopic	36,534
	EndoFM	3,662,705
	Ours (private)	6,501,878
Total in pretraining phase		10,331,747
Downstream	CVC-Clinic DB	600
	Hyper-Kvasir	10,662

- We assemble a large-scale endoscopic image corpus comprising 6.5 million private frames and 3.5 million publicly available images for self-supervised pretraining of EndoMAE.
- Pretrained EndoMAE is used as feature extractor for downstream image classification and polyp segmentation tasks.
- Pretraining and downstream datasets are completely disjoint.

METHOD

1. Dataset generation:

Automated ROI localization, frame extraction & cropping, frame filtering and quality control.

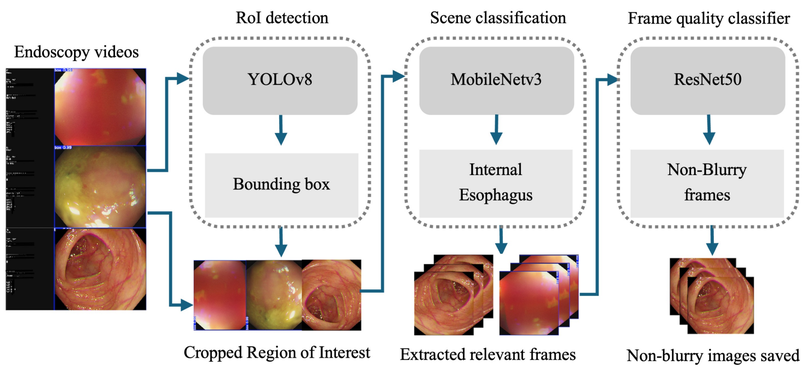

Fig. 1: Overview of the automated dataset generation pipeline.

2. Pretraining EndoMAE with SSL paradigm:

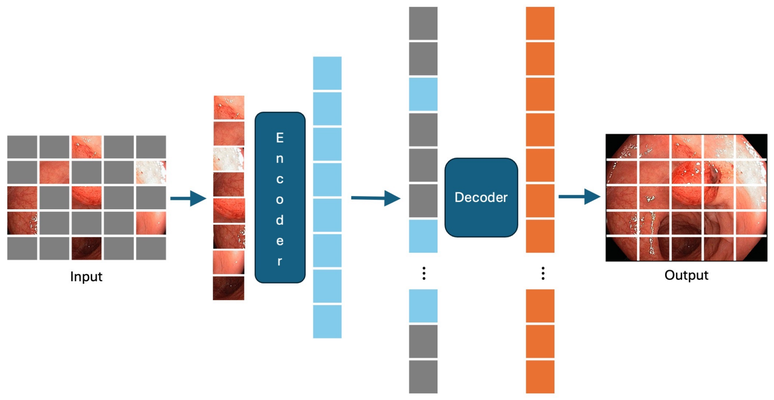

Fig. 2: Self-supervised pretraining pipeline of EndoMAE.

Input endoscopic images are partitioned into patches, and randomly masked. A transformer-based encoder learns latent representations from visible patches, while decoder reconstructs the masked regions using a reconstruction objective.

3. Downstream finetuning:

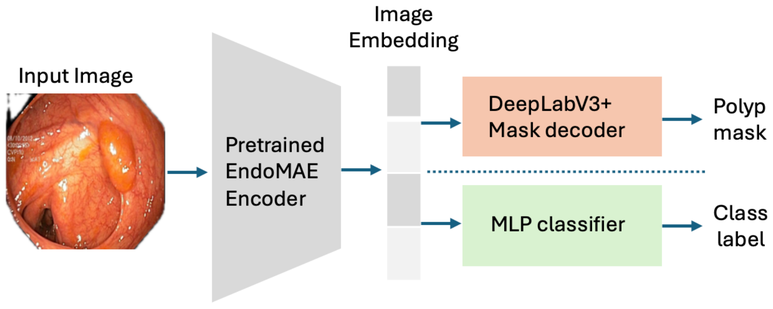

Fig. 3: Overview of downstream finetuning state.

- Pretrained EndoMAE-encoder as feature extractor.
- For classification, a Multi layer perceptron is used.
- For segmentation, deeplabv3+ is adapted for transformer-based EndoMAE as decoder.

RESULTS

EndoMAE feature representation capacity is evaluated against supervised and self-supervised backbones. Finetuning is done in three settings:

(i) Fully frozen encoder (∗).
(ii) Last two blocks are unfrozen (†).
(iii) Full finetuning (‡).

Table 2: Downstream classification result on Hyper-Kvasir.

Model /Backbone	Training Paradigm	F1-score (↑)
ResNet50	Supervised	0.86
C- MixUp	Self-Supervised	0.84
EndoMAE*	Self-Supervised	0.81
EndoMAE†	Self-Supervised	0.87
EndoMAE‡	Self-Supervised	0.89

Note: C-MixUp uses unlabeled Hyper-Kvasir dataset for pre-training, whereas EndoMAE pre-training does not.

Table 3: Downstream segmentation result on CVC-Clinic DB.

Model /Backbone	Training Paradigm	Dice score (↑)
ResNet50	Supervised	0.69
MedSAM	Supervised	0.86
EndoFM	Self-Supervised	0.76
EndoMAE*	Self-Supervised	0.82
EndoMAE†	Self-Supervised	0.84
EndoMAE‡	Self-Supervised	0.86

Note: MedSAM requires prompts (such as points or bounding box to guide segmentation)

CONCLUSION

EndoMAE leverages 10M+ unlabeled endoscopic images for self-supervised pretraining. The pretrained encoder transfers effectively to image classification and polyp segmentation tasks, demonstrating strong cross-dataset generalization and adaptability to diverse endoscopic applications under limited annotation settings.